Staff Summary
Method 2B Application
Renewable Diesel and Renewable Gasoline from Tree Residues
Ensyn Corporation, Tesoro Corporation and Chevron Corporation
(Pathway Codes: RNWG001, RNWG002, RNWD028, and RNWD029)

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Pathway Summary

Ensyn Corporation, Tesoro Corporation, and Chevron Corporation are designing a commercial process to produce renewable gasoline and diesel from the co-processing of Renewable Fuel Oil (also known as bio-oil) obtained via pyrolysis of tree residues. Tree residues essentially consist of sawdust, the woody debris generated during milling operations of trees harvested from managed forests. Tree residues will be collected in the area of Renfrew County, Ontario, Canada and processed into Renewable Fuel Oil (RFO) at a facility in Renfrew, Ontario, Canada. RFO will be transported to California for co-processing in a Fluid Catalytic Cracker (FCC) unit in a refinery into finished renewable gasoline and diesel.

Ensyn Corporation, Tesoro Corporation and Chevron Corporation (hereafter jointly referred to as applicant) expect to produce a combined total of 2 million gallons per year of renewable diesel and gasoline. The renewable gasoline and diesel products meet ASTM specifications and can be blended with ethanol and biodiesel.

The applicant seeks to register four pathways for renewable diesel and gasoline. The applicant requests carbon intensities (CIs) of 20.12 and 25.03 g CO_2e/MJ for renewable gasoline via rail and truck transport, respectively. Similarly, the applicant requests prospective CIs of 21.67 and 25.58 g CO_2e/MJ for renewable diesel via rail and truck transport, respectively. The CI values for these pathways are based on lifecycle analysis conducted using a modified version of the CA-GREET model.

Carbon Intensity of Renewable Gasoline and Renewable Diesel

The proposed pathways are based on the pyrolysis of tree residues into RFO which is then transported to a refinery in California. In the refinery, the RFO is co-processed with Vacuum Gas Oil (VGO) derived from fossil crude for the production of gasoline and diesel. The applicants are requesting the consideration of proportional amounts of renewable gasoline and diesel to be produced from the biogenic feedstock (RFO).

Renewable diesel and gasoline are modeled using a modified version of the CA-GREET model, with inputs modified to reflect process-specific operating scenarios and intermediate results aggregated separately. Well-to-tank greenhouse gas emissions (WTT) are calculated based on energy requirements. The WTT include residue collection, transport to a Renewable Fuel Oil (RFO) production facility, RFO production, RFO transport to a refinery in California, RFO co-processing in the refinery and final transport of finished renewable gasoline and renewable diesel to distribution centers (T&D) in California.

Although tree residues may otherwise be loaded and transported for landfill disposal, the energy use and corresponding greenhouse gas emissions associated with this activity is included in the lifecycle analysis of this pathway. Char and non-condensable gas ("Fuel Gas") are produced as co-product during pyrolysis. Co-products are burned to produce power required by the pre-process and process equipment that produces the RFO. While the CO_2 emissions from this combustion are biogenic, emissions from CH_4 and N_2O were considered in the lifecycle analysis.

Table 1 provides estimated CI values for the four pathways being requested in this application.

Table 1. Proposed Lookup Table Entries

Fuel	Pathway Identifier	Pathway Description*	Carbon Intensity Values (gCO2e./MJ)		
			Direct CI	Indirect LUC	Total
Renewable Gasoline	RNWG001	Method 2B*: Tree residue to renewable gasoline via co-processing of bio oil. Bio oil transport by rail	20.12	0	20.12
Renewable Gasoline	RNWG002	Method 2B*: Tree residue to renewable gasoline via co-processing of bio oil. Bio oil transport by truck.	25.03	0	25.03
Renewable Diesel	RNWD028	Method 2B*: Tree residue to renewable diesel via co-processing of bio oil. Bio oil transport by rail.	21.67	0	21.67
Renewable Diesel	RNWD029	Method 2B*: Tree residue to renewable diesel via co-processing of bio oil. Bio oil transport by truck.	25.58	0	25.58

* Conditions Apply

Operating Conditions

Because the proposed co-processing operation has not commenced commercial operation yet, staff cannot determine actual inputs for the commercial production

of renewable diesel and gasoline production. Staff therefore, imposes constraints on production of renewable diesel and gasoline for these prospective pathways, which are listed below. The applicant shall not use these prospective CIs for purposes of claiming credits unless evidence of actual production along with supporting data is provided and updated provisional CIs are certified for these pathways (which will entail one quarter of commercial production data).

- The applicant shall provide evidence that assumptions regarding pyrolysis yields and energy requirements represent the actual operating conditions. The applicant shall provide quarterly receipts for fuel consumption at the pyrolysis facility including natural gas, electricity, and other energy consumption for two years, reported on a quarterly basis.
- The applicant shall provide quarterly receipts for fuel consumption at the co-processing facility including natural gas, electricity, and other energy consumption for two years, reported on a quarterly basis.
- The applicant shall provide quarterly receipts for two years, reported on a quarterly basis to support quantity of biomass used and renewable gasoline and diesel produced.
- As is the case with petroleum in a refinery, literature related to co-processing indicates that not all the bio-oil in the feed is transferred over to the finished gasoline or diesel streams. The applicant shall therefore provide evidence of biogenic component in finished gasoline and diesel fuel. That evidence shall include results from acceptable techniques (e.g., C14) or other comparable techniques that establish that the reported volumes eligible to earn LCFS credits contain biogenic, and therefore renewable carbon.
- In addition, the applicant shall provide evidence of the percentages of finished gasoline and diesel fuel that is attributed to the introduction and addition of RFO using the standard and reliable mass balance techniques that establish gasoline and diesel yields from petroleum feedstocks.

Staff Analysis and Recommendation

ARB staff has reviewed the application and has replicated the carbon intensity values calculated for each of the four pathways using the provided CA-GREET model. The applicant provided documentation for the energy inputs for RFO production based on process modeling. Based on a preliminary assessment using submitted process information, staff has estimated carbon intensity values for each of the four pathways as proposed by the applicant. Staff believes that the prospective CI values indicated here are a reasonable representation of the carbon intensities of renewable diesel and gasoline. Because the applicant provided only limited process data, staff recommends that the Method 2B application be approved as prospective.

Fuels with prospective CIs are not eligible to claim credits under the LCFS. To claim credits, the applicant must provide one quarter of operational data once

commercial production has commenced. ARB will then complete an updated lifecycle analysis and make necessary adjustments to the originally certified prospective CIs if warranted and approve a provisional CI for each of the pathways being considered in this application. To confirm compliance with updated operating conditions, the Executive Officer may reevaluate any aspect of the review at any time and revise the certification to reflect new information. At any time after certification, the Executive Officer may increase the CI values upon determination that the provisional CIs underestimate fuel life carbon intensity. (Cal. Code Regs. tit. 17, § 95486, subd. (e)(3)(K) (original LCFS); Cal. Code Regs. tit. 17, § 95488, subd. (c)(5)(L) (from January 1, 2016).)